Big-Op and Little-op Notation

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Introduction

Asymptotic statistics constantly deals with sequences that are "approximately" of a given size as $n \to \infty$. Big-O and little-o notation from calculus handle deterministic sequences; their stochastic counterparts – big-O_p and little-o_p – handle random ones. These symbols compress pages of tedious bookkeeping into a few letters and are essential reading for any modern statistics paper.

Prerequisites

The reader should understand convergence in probability and deterministic big-O, little-o notation.

Theory

Big-O_p (stochastic boundedness). A sequence Y_n is $O_p(r_n)$ if for every $\varepsilon > 0$ there exist M, N such that

$$P(|Y_n/r_n| \leq M) \geq 1 - \varepsilon \quad \text{for all } n \geq N.$$

Equivalently, Y_n/r_n is stochastically bounded – it does not grow without bound in probability.

Little-o_p (negligibility). A sequence Y_n is $o_p(r_n)$ if $Y_n/r_n \xrightarrow{P} 0$.

Standard asymptotic rates.

- $\bar{X}_n - \mu = O_p(n^{-1/2})$ by the CLT.
- $\hat{\sigma}_n^2 - \sigma^2 = O_p(n^{-1/2})$ similarly.
- If $\hat{\theta}_n$ is consistent, $\hat{\theta}_n - \theta = o_p(1)$.
- In regression, $\hat{\beta}_n - \beta = O_p(n^{-1/2})$ for OLS under standard conditions.

Algebraic rules.

- $O_p(1) + O_p(1) = O_p(1)$ (sum of bounded remains bounded).
- $o_p(1) + o_p(1) = o_p(1)$.
- $O_p(a_n) \cdot O_p(b_n) = O_p(a_n b_n)$.
- $O_p(1) \cdot o_p(1) = o_p(1)$ (bounded times negligible is negligible).

These rules let us manipulate asymptotic expressions without repeatedly invoking convergence definitions.

Delta-method expansion. If $\hat{\theta}_n - \theta = O_p(n^{-1/2})$ and g is smooth,

$$g(\hat{\theta}_n) - g(\theta) = g'(\theta)(\hat{\theta}_n - \theta) + O_p(n^{-1}).$$

The leading term is what the delta method keeps; the $O_p(n^{-1})$ remainder is negligible compared to the $O_p(n^{-1/2})$ leading term, so the asymptotic distribution is determined by the first-order expansion alone.

Assumptions

Stochastic order symbols are defined under the usual probability-space setup; the concepts are distributional, not almost-sure, so they work even when pointwise behaviour is complicated.

R Implementation

```r
set.seed(2026)

n_vals <- round(10^seq(1, 4, by = 0.5))
reps <- 1000

sim <- sapply(n_vals, function(n) {
  dev <- replicate(reps, abs(mean(rnorm(n)) - 0))
  mean(dev * sqrt(n))
})

# If xbar - mu = O_p(n^-1/2), sqrt(n) * (xbar - mu) should be bounded
data.frame(n = n_vals, mean_abs_sqrt_n = sim)
```

We check empirically that $\sqrt{n}(\bar{X}_n - \mu)$ is stochastically bounded: the mean of its absolute value should stabilise as n grows.

Output & Results

```
      n  mean_abs_sqrt_n
1    10        0.829
2    32        0.808
3   100        0.794
4   316        0.801
5  1000        0.797
6  3162        0.798
7 10000        0.797
```

The product stabilises near $\sqrt{2/\pi} \approx 0.798$, the mean absolute value of a standard normal – exactly as the $O_p(n^{-1/2})$ rate predicts.

Interpretation

When a paper writes "the estimator is $\sqrt{n}$-consistent" or "the error is $O_p(n^{-1/2})$", it is making a formal statement about the rate at which the estimator approaches the truth. Different estimators can have different rates (parametric vs. non-parametric, standard vs. super-efficient), and these rates directly affect sample-size planning.

Practical Tips

- $\sqrt{n}$-consistent is the gold standard for parametric estimators; non-parametric estimators often have slower rates like $n^{-1/3}$ or $n^{-2/5}$.
- In bias-variance decompositions, knowing the rate of each term tells you which dominates at what sample size.
- When the leading term in a Taylor expansion is $O_p(n^{-1/2})$ and the remainder is $O_p(n^{-1})$, the expansion is consistent and the delta method applies.
- Do not confuse "rate" with "variance" – the rate is about the magnitude of the random variable, not its specific distribution.
- For complex estimators (profile likelihood, semi-parametric), establishing the O_p rate is often the first technical step in any asymptotic derivation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr